

Voxa+ — Subscriber Churn: Root-Cause Diagnosis

The exhaustive record behind the board summary · dual-track (DuckDB SQL + pandas)

Voxa+ — Subscriber Churn: Root-Cause Diagnosis

Analyst deep-dive · prepared for the leadership team · all figures dual-computed in DuckDB SQL and pandas with a parity check on every result (`../parity/parity_report.md` — 14/14 pass).

0. The question and the answer

Question. Gross monthly subscriber churn nearly doubled starting at window-month 34 (Jun 2026) and has stayed elevated. What caused it?

Answer. Three factors that overlapped in time. None explains the spike on its own.

	Factor	Role	Markets
1	An app "v3" connected-TV playback regression (window-month 32)	acute trigger, biggest contributor	all; weakest in the US
2	The Brazil Standard/Premium price increase (window-month 32), landing on an already-disengaged base	amplifier	Brazil only
3	A months-old Mexico cohort-quality drag (newer Mexican subscribers retain ~8 pp worse)	pre-existing drain	Mexico only

Payments and involuntary churn were tested and are **not** involved (§6).

1. The spike — size, persistence, shape (*findings F01, F02*)

Gross monthly churn stepped from a **4.1 %** pre-shift baseline (window-months 25–33) to **6.8 %** across the three elevated months — a **1.66×** increase, and **90 % of it is voluntary**. It has not receded: the three months read 6.85 %, 6.58 %, 7.02 %.

Splitting by market shows the company-wide number is a composite of three different shapes:

Market	Pre-shift	Peak	Behaviour
Brazil	3.4 %	7.0 %	flat, then a clean +3.0 pp step at window-month 34 — acute
Mexico	5.4 %	8.1 %	drifting up from ~window-month 29 (+0.6 pp), then a further step — a problem that got worse
United States	4.1 %	4.4 %	essentially unchanged — the control

A US that barely moves already rules out any explanation that would hit every market equally.

2. Who is leaving, and why now (*findings F03, F04*)

The lift is broad, not narrow. Peak-vs-baseline churn is up by a strikingly even $\sim +2.7$ pp across almost every tenure band, plan tier and acquisition channel. Two exceptions carry information:

- **Annual plans lifted only +1.9 pp** (vs +2.7 pp monthly) — locked-in subscribers were partly shielded.
- **Long-tenure subscribers (25+ months) were hit as hard as brand-new ones** (+3.2 pp) — a cause confined to weak new cohorts could not do that.

A single narrow cause does not produce an even cross-segment lift. This is the first quantitative sign of **multiple overlapping factors**.

The extra churn is engagement-led. For subscribers who cancelled during the elevated months, streamed hours were already sliding beforehand — ~ 37 h two months out, ~ 36 h the month before — whereas pre-shift churners left from a flat ~ 43 h. Value erodes first, the cancellation follows. That points at what suppressed viewing just before the spike (§3), and it explains why a price change bites harder now (§4): people were already getting less.

3. Factor 1 — the connected-TV app regression (*finding F05*)

The Voxa+ "v3" app shipped device-family by device-family, reaching connected-TV last. Connected-TV streaming quality is flat through window-month 31, then:

window-month	31	32	33–36
CTV video-start-failure rate	1.1 %	3.0 %	~3.4 %
CTV playback-error rate	0.6 %	1.4 %	~1.6 %

Mobile and web are unchanged throughout. The connected-TV rates **triple at window-month 32 — one month before the churn step**.

The churn consequence is specific to the affected subscribers:

	Pre-shift	Peak
Not CTV-heavy	4.1 %	5.6 %
CTV-heavy (primary device = smart-TV / stick / console)	4.1 %	8.4 %

Before the shift the two groups churned **identically**. After the CTV rollout, CTV-heavy churn **doubled**. The one-month lag between the quality drop and the churn step is a value → decision delay of roughly one billing cycle. This is the **acute trigger and the largest single contributor**, and — because it touches every market — it is why the company-wide number looks global.

4. Factor 2 — the Brazil price increase, and why it bit (*finding F06*)

Brazil raised Standard and Premium list prices at window-month 32, migrating existing members on their next renewal. Split Brazil Std/Prem subscribers by whether they were engaged or disengaged (below the healthy-engagement line) in the month:

Brazil Std/Prem	Churn before hike	Churn after hike	Lift
Engaged	~0 %	~0 %	≈ 0 pp
Disengaged	23.9 %	42.2 %	+18.3 pp

The price increase had **no measurable effect on engaged subscribers** and a very large effect on **already-disengaged** ones. Because Factor 1 pushed a wave of Brazil subscribers into the disengaged bucket in window-months 32–33, the migration then landed on a larger disengaged pool than it otherwise would have. Price alone did not move the base; **price × prior disengagement produced the Brazil step**. Brazil-only.

5. Factor 3 — Mexico's pre-existing cohort-quality drag (*finding F07*)

Month-3 retention by market and cohort era:

Market	Pre-shift cohorts	Post-shift cohorts (acquired m24–30)
Brazil	~82.8 %	~80 % (–3 pp)
Mexico	~81.8 %	~73 % (–8–9 pp)
United States	~82.8 %	~81 % (–2 pp)

Every market's newer cohorts retain a little worse; **Mexico's are ~8–9 pp worse at month 3** — three times the drop elsewhere. It is **not channel-specific** (within post-shift Mexico, low-quality-channel cohorts retain about the same as the rest), so this is not "the channel mix kept degrading" — it is **Mexican conditions for newly-acquired subscribers deteriorating across the board** from ~window-month 24. This drag was already lifting Mexican churn ~0.6 pp before the visible spike and is still underneath it.

6. What it is *not* — payments (*finding F08*)

Market	First-attempt payment-failure rate (pre → peak)	Dunning recovery (pre → peak)	Involuntary churn (pre → peak)
Brazil	7.6 % → 7.7 %	51 % → 52 %	0.43 % → 0.41 %
Mexico	18.8 % → 18.2 %	37 % → 37 %	1.25 % → 1.30 %
United States	8.0 % → 7.8 %	58 % → 57 %	0.31 % → 0.32 %

Nothing moved. Combined with the 90 % voluntary share (§1), payments are **ruled out** as a driver of the spike. Mexico's payment economics are structurally poor but that is a standing cost, not a new one.

7. Impact, and the decomposition (*findings F09, F10*)

Impact. The churn step cuts the implied average subscriber lifetime from ~24 months to ~15 — a ~40 % reduction in lifetime value. In the three elevated months, ~5,400 subscribers left over and above the pre-shift rate; at ~\$7.5 ARPU that is on the order of \$0.5–0.6 M of lifetime recurring revenue already forgone, compounding monthly. A 40 % lower LTV also cuts the allowable CAC by ~40 %, so the growth plan no longer pencils.

Decomposition. Tag every peak-month voluntary churn with the three factor flags:

Market	Peak voluntary churn	F1 (CTV)	F2 (price×diseng)	F3 (low-q young)	none
Brazil	6,629	54 %	63 %	12 %	15 %
Mexico	4,017	53 %	0 %	23 %	36 %
United States	1,597	52 %	0 %	20 %	39 %

In Brazil, 85 % of the elevated churn carries Factor 1 or Factor 2. In Mexico, ~64 % carries Factor 1 or Factor 3. The US "39 % none" is just its unchanged ~4 % baseline — the control. The flags overlap, so these are *coverage*, not an independent split; the causal weight sits in the **rate** comparisons (§3–§5), which are clean.

8. Recommendations (priority order)

1. **Fix the connected-TV player (Sev-1).** Roll back / hotfix v3 on smart-TV and streaming-stick; verify video-start-failure returns to ~1 %. Save-campaign CTV-heavy subscribers whose viewing fell after window-month 32.
 2. **Pause the Brazil price migration for low-engagement Standard/Premium subscribers.** Offer a retention path (hold price / downgrade / pause); re-test elasticity after the CTV fix.
 3. **Audit Mexico acquisition and onboarding since mid-2025.** Cap spend on the weakest sources until month-3 retention recovers toward ~82 %.
 4. **Instrument for next time.** Connected-TV QoE on the executive dashboard, a leading "engagement health" trend, and price-change elasticity by engagement band.
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How this was produced

`python data_analysis/run_analysis.py` runs all ten questions twice — DuckDB SQL over the curated Parquet and pandas over the same files — and asserts the two agree within tolerance before any number is used. SQL is saved under `data_analysis/sql/`, agreed results under `data_analysis/results/`, and the parity log at `data_analysis/parity/parity_report.md`. Per-problem method, evidence and limitations are in `data_analysis/findings/F01 – F10`.

Appendix — the ten findings in full

Each numbered problem, with its hypothesis, both-tracks method, evidence table, conclusion, limitations and recommendation — exactly as recorded during the analysis.

F01 — The spike: size, split, and whether it is still going

Context. Leadership reported that monthly churn "nearly doubled" ~3 months ago. **Question.** By how much did gross monthly churn rise, is it voluntary or involuntary, and has it plateaued, kept climbing, or started to recede?

Hypothesis. Gross monthly churn roughly doubled at window-month 34 and has stayed elevated; the increase is predominantly voluntary (member-initiated), not payment-driven.

Method (both tracks). From `fact_subscription_month`, per `month_idx`: paid-active count, voluntary and involuntary churn counts. Gross monthly churn = (voluntary + involuntary) ÷ active at the start of the month (prior-month active). Pre-shift baseline = mean over `month_idx` 24–32; peak = mean over 33–35. DuckDB `q1_churn_trend` and the pandas equivalent agree exactly (`parity/parity_report.md`).

Evidence (`results/q1_summary.csv`, `results/q1_churn_trend.csv`).

	Value
Pre-shift blended gross monthly churn	4.11 %
Peak blended (window-months 34–36)	6.82 %
Ratio	1.66×
Month-by-month at the peak	34 = 6.85 % · 35 = 6.58 % · 36 = 7.02 %
Voluntary share of peak churn	90.4 %

Conclusion. Gross monthly churn stepped from ~4.1 % to ~6.8 % — a **1.66×** increase, fairly described as "nearly doubled." It has **not** receded: the three elevated months are 6.85 %, 6.58 %, 7.02 % — flat-to-rising. The rise is **overwhelmingly voluntary** (people are choosing to leave), which points the investigation at product, price and value, not at billing mechanics (that is tested and ruled out in `F08`).

Limitations. Blended monthly-average rate; a single unusually large acquisition month can move the denominator. Reactivations are excluded from the churn numerator (they are tracked separately).

Recommendation. Treat this as an active, unresolved leak — plan on the ~4 % baseline not being restored on its own. Size the response to a sustained ~2.7 pp of excess monthly churn

(see F09 for the subscriber and revenue impact).

F02 — Is it global, or concentrated in one market?

Question. Did churn rise the same way in Brazil, Mexico and the United States, or is the company-wide number hiding different stories per market?

Hypothesis. The spike is not uniform: at least one market steps up sharply at window-month 34 while another was already drifting up beforehand, and a third is roughly stable.

Method (both tracks). `sm view ( fact_subscription_month joined to dim_subscriber )`. Per market per `month_idx`: gross monthly churn (as in F01). Two derived quantities per market: the **drift** (mean churn over months 28–32 minus mean over 24–27) and the **step** (month 33 minus month 32). `q2_churn_by_market / q2_market_summary`; SQL and pandas agree exactly.

Evidence (`results/q2_market_summary.csv`).

Market	Pre-shift	Peak	Drift (m28–32 vs m24–27)	Step (m32 → m33)
Brazil	3.4 %	7.0 %	+0.2 pp (none)	+3.0 pp
Mexico	5.4 %	8.1 %	+0.6 pp	+2.3 pp
United States	4.1 %	4.4 %	−0.3 pp	+0.7 pp

Conclusion. Three different shapes:

- **Brazil** — flat at ~3.4 % through window-month 33, then a clean **+3.0 pp step** to ~7.0 %. An acute event.
- **Mexico** — already the highest market, and **drifting upward from ~window-month 28** (before anyone was looking), then a further step to ~8.1 %. A pre-existing problem that got worse.
- **United States** — essentially unchanged (4.1 % → 4.4 %). The control.

The company-wide "nearly doubled" is a Brazil step **plus** a Mexico ramp; the US shows the shift is not a platform-wide, every-market phenomenon. That already rules out explanations that would hit all three markets equally.

Limitations. Mexico and the US have shorter histories; the US in particular has a smaller base and noisier monthly rates.

Recommendation. Run two workstreams, not one: an acute-incident review focused on Brazil (what changed at window-month 32–33), and a separate look at why Mexico has been deteriorating since mid-2025. F05 – F07 identify the specific causes.

F03 — Which cohorts, tenures, plans and channels are churning?

Question. Is the extra churn concentrated in one segment (new sign-ups, a plan tier, a weak channel), or is it broad-based?

Hypothesis. If a single cause dominated we would expect the churn lift to land mostly in one segment. A broad, roughly even lift across segments would instead point to several overlapping causes.

Method (both tracks). `sm` view. For each cut — plan tier, tenure band (0–1, 2–6, 7–12, 13–24, 25+ months), acquisition channel, billing period — gross monthly churn averaged over the pre-shift baseline (24–32) and the peak (33–35), and the lift in percentage points. `q3_cohort_cuts` ; SQL and pandas agree exactly.

Evidence (`results/q3_cohort_cuts.csv` , lift = peak – pre-shift).

Cut	Segment	Pre-shift	Peak	Lift
Tenure	25+ months	3.4 %	6.6 %	+3.2 pp
Tenure	13–24 months	2.7 %	5.9 %	+3.1 pp
Tenure	0–1 months	7.6 %	10.7 %	+3.1 pp
Tenure	2–6 months	4.5 %	7.1 %	+2.6 pp
Tier	Básico	4.8 %	7.6 %	+2.8 pp
Tier	Standard	4.2 %	6.9 %	+2.7 pp
Tier	Premium	3.0 %	5.7 %	+2.7 pp
Channel	paid_search / paid_social / organic / direct	~4.0 %	~6.8 %	+2.7–2.8 pp
Billing	monthly	4.1 %	6.8 %	+2.7 pp
Billing	annual	5.9 %	7.8 %	+1.9 pp

Conclusion. The lift is **broad and remarkably even** — roughly +2.7 pp on almost every tenure band, tier and channel. Two exceptions are informative:

1. **Annual plans lifted only +1.9 pp** vs +2.7 pp for monthly — locked-in subscribers were partly shielded (they cannot act until renewal, and are stickier by selection).
2. **Long-tenure subscribers (25+ months) were hit as hard as brand-new ones.** A cause confined to bad new cohorts would not move the 13–24 and 25+ bands by +3 pp.

A single narrow cause does not produce this pattern. The even, cross-segment lift is the first quantitative sign that **multiple factors are acting at once**, each touching a different slice of the base.

Limitations. Acquisition channel is a sampled attribute (`docs/08` limitation 1), so channel-level rates are directional. Tenure band uses `tenure_months` which resets on reactivation.

Recommendation. Do not chase a single segment. The next findings test the specific mechanisms — app quality by device (F05), the Brazil price change (F06), and Mexico acquisition/cohort quality (F07) — and F10 shows how they add up.

F04 — Did viewing behaviour drop *before* people cancelled?

Question. For subscribers who cancelled during the elevated months, was their viewing already falling in the weeks before, or did they leave from a healthy level of use?

Hypothesis. Peak-period churners show a decline in streamed hours in the one or two months before cancellation that pre-shift churners do not — i.e. disengagement leads the cancellation, rather than the cancellation being a bolt from the blue.

Method (both tracks). For every voluntary churn in `fact_subscription_month`, join `fact_engagement_month` to read streamed hours in the churn month `M`, `M-1` and `M-2`, and whether the subscriber was below the healthy-engagement threshold (`< 5 h`) in `M-1`. Averages split by whether the churn month is pre-shift (24–32) or peak (33–35).

`q4_engagement_before_churn`; SQL and pandas agree exactly.

Evidence (`results/q4_engagement_before_churn.csv`).

Churn cohort	Churners	Avg hours M-2	Avg hours M-1	Below healthy in M-1
Pre-shift (m24–32)	17,146	43.2 h	43.6 h	11.5 %
Peak (m33–35)	12,243	37.3 h	35.7 h	12.7 %

Conclusion. Pre-shift churners left from a roughly flat ~43 h/month — their exit was not preceded by a viewing decline. **Peak-period churners were already sliding:** ~37 h two months out, ~36 h the month before they cancelled, and a higher share (12.7 % vs 11.5 %) had fallen below the healthy-engagement line. The extra churn is **engagement-led** — value erodes first, the cancellation follows. That tells us to look for what suppressed viewing in the months just before the spike (F05), and it explains why a price change would bite harder now than before (F06): people were already getting less for the money.

Limitations. `M` (churn-month) engagement is sparse — many subscribers do not stream in the month they cancel — so the analysis leans on `M-2 / M-1`. Engagement is measured at the subscriber-month grain from `fact_engagement_month`.

Recommendation. Add a leading "engagement health" watch to the retention dashboard (share of the base below the healthy line, and its 2-month trend) so the next value shock is visible before it turns into cancellations.

F05 — The connected-TV app-quality regression (*Factor 1*)

Question. Did streaming quality degrade on any device family in the months before the spike, and did the subscribers who rely on that device churn more?

Hypothesis. The Voxa+ "v3" app rollout — which reached connected-TV devices last, around window-month 32–33 — introduced a playback regression on smart-TV / streaming-stick / console. Subscribers who watch mostly on those devices lost quality, then value, then churned a month later.

Method (both tracks). 1. `fact_engagement_device_month` joined to `dim_device`: video-start-failure rate and playback-error rate (failures ÷ play starts) by **device class** by `month_idx`. 2. `sm` view: gross monthly churn for **CTV-heavy** subscribers (primary device in smart_tv / streaming_stick / console) vs the rest, pre-shift vs peak. `q5_qoe_by_device`, `q5_ctv_heavy_churn`; SQL and pandas agree exactly.

Evidence.

Connected-TV quality (`results/q5_qoe_by_device.csv`):

window-month	31	32	33	34	35	36
CTV video-start-failure rate	1.09 %	3.05 %	3.41 %	3.42 %	—	3.44 %
CTV playback-error rate	0.59 %	1.43 %	1.58 %	1.55 %	—	1.57 %

Mobile and web rates are flat at ~1.1 % / ~0.6 % throughout. The CTV rates **triple at window-month 32** — exactly one month before the churn step at window-month 34.

CTV-heavy vs rest — gross monthly churn (`results/q5_ctv_heavy_churn.csv`):

Segment	Pre-shift	Peak
Not CTV-heavy	4.09 %	5.65 %
CTV-heavy	4.14 %	8.42 %

Conclusion. Before the shift, CTV-heavy and non-CTV subscribers churned at the **same** rate (~4.1 %). After the v3 CTV rollout, CTV-heavy churn **doubled to 8.4 %** while the rest rose only modestly. The one-month lag between the quality drop (window-month 32) and the churn step (window-month 34, i.e. `month_idx` 33) is consistent with a value → decision delay of roughly one billing cycle. This is **Factor 1**: an app regression on the big-screen experience, hitting the heaviest-viewing part of the base. It is present in every market but weakest in the US (`F02`), which had a smoother CTV rollout.

Limitations. "CTV-heavy" is proxied by the subscriber's primary device family (a `dim_subscriber` attribute), not a live per-month device mix. Playback telemetry is synthetic and aggregated at `fact_engagement_device_month`.

Recommendation. Treat the v3 connected-TV player as a Sev-1: roll back or hotfix the start-up / playback path on smart-TV and streaming-stick, and confirm the video-start- failure rate returns

to ~1 %. Run a save campaign for CTV-heavy subscribers whose viewing dropped after window-month 32.

F06 — The Brazil price increase, and why it bit (*Factor 2*)

Question. Brazil raised Standard and Premium list prices at window-month 32 (migrating existing members on their next renewal). Did that trigger the Brazil step in churn — and if so, who left?

Hypothesis. The price increase on its own is not enough to explain a +3 pp step (in-place price rises usually cause a small bump). It bit because it landed on a base that was **already disengaged** — so the churn lift is concentrated in Brazil Standard/Premium subscribers who were below the healthy-engagement line, not in engaged ones.

Method (both tracks). `sm` view restricted to Brazil, Standard/Premium tiers. Each subscriber-month tagged **disengaged** if below the healthy-engagement threshold that month (`fact_engagement_month` ; missing → disengaged). Gross monthly churn by month for the engaged and disengaged segments, compared before the hike (months 26–29) and after (months 31–35). `q6_br_price_engaged_split` , `q6_price_summary` ; SQL and pandas agree exactly.

Evidence (`results/q6_price_summary.csv`).

Brazil Std/Prem segment	Churn before hike (m26–29)	Churn after hike (m31–35)	Lift
Engaged	0.0 % (<i>rounds to zero — very low</i>)	0.0 %	≈ 0 pp
Disengaged	23.9 %	42.2 %	+18.3 pp

Conclusion. The price increase had **no measurable effect on engaged Brazil subscribers** and a very large effect on **already-disengaged** ones (+18 pp on a segment that was already churning at ~24 %/month). The disengaged Std/Prem segment is small, but the interaction is exactly the amplifier the hypothesis predicted: **price alone did not move engaged users; price × prior disengagement produced the Brazil step**. Because `F05` (the CTV regression) pushed a wave of Brazil subscribers into the disengaged bucket in window-months 32–33, the price migration then landed on a bigger disengaged pool than it otherwise would have. This is **Factor 2**, and it is Brazil-only.

Limitations. The engaged-segment rate is so low it rounds to zero at the reported precision; the signal is entirely in the disengaged segment. Migration timing is approximate (existing members move on their renewal date, spread across months 32–36).

Recommendation. For Brazil Standard/Premium, pause or slow the remaining price migrations for subscribers flagged as low-engagement; offer them a retention path (a month of the prior

price, a downgrade to Básico, or a pause) rather than a renewal at the new price. Re-test the elasticity once the CTV fix (F05) has restored engagement.

F07 — Mexico: a pre-existing acquisition / cohort-quality drag (Factor 3)

Question. Mexico's churn was already the highest and was drifting up before the spike (F02). Is that a cohort-quality problem — are the subscribers Mexico has been acquiring since mid-2025 simply retaining worse?

Hypothesis. Mexican cohorts acquired from ~window-month 24 onward retain materially worse than earlier Mexican cohorts, and worse than the equivalent drop in Brazil or the US — a slow-building drag that predates, and then compounds with, the app regression.

Method (both tracks). 1. `dim_subscriber` + `dim_channel` : share of each cohort-month's new subscribers coming through low-quality channels (affiliate + partner bundle), by market. 2. First-spell start month per subscriber (`fact_subscription_month`), and whether the subscriber was still active at tenure month 3. **M3 retention** by market × channel quality × cohort era: *pre-shift cohorts* = first paid month 12–23, *post-shift cohorts* = 24–30. `q7_lowq_share_by_cohort` , `q7_m3_retention_by_channel_quality` ; SQL and pandas agree exactly.

Evidence (`results/q7_m3_retention_by_channel_quality.csv`).

Market	Cohort era	M3 retention (low-quality ch.)	M3 retention (other ch.)
Brazil	pre-shift	82.9 %	82.7 %
Brazil	post-shift	80.2 % (–2.7 pp)	79.7 % (–3.0 pp)
Mexico	pre-shift	81.7 %	81.9 %
Mexico	post-shift	74.0 % (–7.7 pp)	72.9 % (–9.0 pp)
US	pre-shift	83.0 %	82.5 %
US	post-shift	81.0 % (–2.0 pp)	80.9 % (–1.6 pp)

Low-quality-channel share of Mexican new subscribers sat at ~46 % **throughout** window-months 24–36 (`results/q7_lowq_share_by_cohort.csv`) — high, but not *rising*.

Conclusion. Every market's newer cohorts retain a little worse, but **Mexico's are ~8–9 pp worse at month 3** — three times the Brazil or US drop. Notably it is **not channel-specific**: within post-shift Mexico, low-quality-channel cohorts (74.0 %) retain about the same as everyone else (72.9 %). So this is not "the channel mix kept getting worse" — it is **Mexican market conditions for newly-acquired subscribers deteriorating across the board** from ~window-month 24, on top of an already-high involuntary base (F08). That is **Factor 3**: a months-old, Mexico-specific cohort-quality drag that was lifting Mexican churn ~0.6 pp before window-month 33 (F02) and is still there underneath the spike.

Limitations. Acquisition channel is sampled (docs/08 limitation 1), so the low-vs-other split is directional; the era-level M3 drop is the robust signal. Cohort eras are bounded so every cohort has ≥ 3 months of observable tenure inside the window.

Recommendation. Audit Mexican acquisition since mid-2025 — creative, offers, bundle terms, onboarding — and the first-90-day experience specifically for Mexico. Until M3 retention recovers toward $\sim 82\%$, cap spend on the weakest Mexican sources and rebalance toward the channels whose cohorts still retain.

F08 — Payment failures and involuntary churn: tested and ruled out

Question. Did payment authorisation, dunning recovery, or involuntary churn deteriorate at the shift — i.e. are people being *dropped* rather than *leaving*?

Hypothesis (to falsify). The spike is (partly) involuntary: failed renewals rising or dunning recovery falling, especially in Mexico with its OXXO / carrier-billing mix.

Method (both tracks). `fact_billing_attempt` : first-attempt payment-failure rate (failed $\div$ non-dunning attempts) and dunning-recovery rate (recovered $\div$ dunning attempts), by market, pre-shift vs peak. `sm` view: involuntary churn rate by market, pre-shift vs peak. `q8_payments` , `q8_involuntary_by_market` ; SQL and pandas agree exactly.

Evidence (`results/q8_payments.csv` , `results/q8_involuntary_by_market.csv`).

Market	Payment-failure rate (pre $\rightarrow$ peak)	Dunning recovery (pre $\rightarrow$ peak)	Involuntary churn (pre $\rightarrow$ peak)
Brazil	7.6 % $\rightarrow$ 7.7 %	50.7 % $\rightarrow$ 52.0 %	0.43 % $\rightarrow$ 0.41 %
Mexico	18.8 % $\rightarrow$ 18.2 %	37.0 % $\rightarrow$ 36.8 %	1.25 % $\rightarrow$ 1.30 %
United States	8.0 % $\rightarrow$ 7.8 %	57.9 % $\rightarrow$ 56.9 %	0.31 % $\rightarrow$ 0.32 %

Conclusion. Nothing moved. Payment-failure rates, dunning recovery and involuntary churn are flat within noise across every market from pre-shift to peak. Mexico's payment economics are structurally worse (18 % first-attempt failure, 37 % recovery, 1.3 % involuntary churn) — a standing cost, not a new one. Combined with F01 (90 % of peak churn is voluntary), this **rules payments out as a driver of the spike**. The elevated churn is people deciding to leave, not billing dropping them.

Limitations. Payment method mix and authorisation rates are modelled (docs/01 §9, docs/08); the finding is about the *change* over time, which is robust to the absolute calibration.

Recommendation. No action on the spike from here. Separately, Mexico's $\sim 18\%$ payment-failure rate and 37 % dunning recovery are worth a dedicated project — every point of recovery

there is ~0.1 pp off Mexican churn — but that is an efficiency programme, not the answer to the board's question.

F09 — What the elevated churn is costing

Question. Translate the ~2.7 pp of extra monthly churn into subscribers and revenue, so the board can size the response.

Method (both tracks). From `fact_subscription_month`: blended gross monthly churn pre-shift vs peak; implied average lifetime = $1 \div \text{monthly churn}$; ARPU and MRR at the peak. Cumulative **excess churned subscribers** since the shift = Σ over months 33–35 of (actual churned – start-of-month active $\times$ pre-shift churn rate). `q9_financial`, `q9_excess_churn`; SQL and pandas agree exactly.

Evidence (`results/q9_financial.csv`, `results/q9_excess_churn.csv`).

	Pre-shift	Peak
Blended gross monthly churn	4.11 %	6.82 %
Implied average subscriber lifetime	24.3 months	14.7 months
ARPU (USD, at peak)		\$7.51
MRR (USD, at peak)		~\$499 k / month

	Value
LTV erosion (lifetime peak $\div$ lifetime pre – 1)	–39.5 %
Excess subscribers churned in the 3 elevated months vs the pre-shift rate	≈ 5,400

Conclusion. The churn step cuts the average subscriber lifetime from ~24 to ~15 months — a **~40 % reduction in lifetime value per subscriber**. In the three elevated months alone, **~5,400 subscribers left over and above what the old rate would have produced**; at ~\$7.5 ARPU and a ~15-month remaining lifetime, that is on the order of **\$0.5–0.6 M of lifetime recurring revenue already forgone**, and it compounds every month the rate stays elevated. It also breaks the growth maths: a 40 % lower LTV means the allowable CAC falls by ~40 %, so acquisition that was economic at the old churn rate is not at the new one.

Limitations. Lifetime = $1 \div \text{current monthly churn}$ is a standard but simple proxy; it overstates the loss if churn reverts quickly and understates it if the rate keeps climbing. The revenue figure is an order-of-magnitude estimate, not a booked number.

Recommendation. Fund the `F05` connected-TV fix and the `F06` Brazil price-migration pause as revenue-protection, not cost. Freeze or trim spend on the least-economic acquisition sources

(Mexico especially, F07) until LTV recovers, and re-plan net-adds on the ~15-month lifetime, not the ~24-month one, until the rate is back near 4 %.

F10 — Synthesis: the root cause is a combination of three factors

Question. Put F02 – F08 together. What actually caused the spike, and can any single factor account for it?

Method (both tracks). Every **voluntary** churn in the three elevated months (month_idx 33–35) tagged with three flags:

- f1_ctv — subscriber's primary device is connected-TV (exposed to the F05 regression);
- f2_price_diseng — Brazil, Standard/Premium, and disengaged that month (exposed to the F06 interaction);
- f3_lowq_young — acquired through a low-quality channel and ≤ 5 months tenure (the F07 cohort-quality drag).

Share of peak voluntary churn carrying each flag, and the share carrying **none**, by market.
q10_driver_decomposition ; SQL and pandas agree exactly.

Evidence (results/q10_driver_decomposition.csv).

Market	Peak voluntary churn	has F1 (CTV)	has F2 (price×diseng)	has F3 (low-q young)	has none of the three
Brazil	6,629	53.8 %	63.4 %	12.0 %	15.2 %
Mexico	4,017	52.6 %	0 %	22.8 %	36.0 %
United States	1,597	51.5 %	0 %	19.8 %	38.6 %

Read alongside the rate evidence:

- CTV-heavy churn 4.1 % → **8.4 %** vs non-CTV 4.1 % → 5.6 % (F05);
- disengaged Brazil Std/Prem churn **+18 pp** after the price hike, engaged **~0** (F06);
- Mexican post-shift cohorts **–8–9 pp** M3 retention (F07);
- payments and involuntary churn **unchanged** (F08).

Conclusion — the diagnosis.

The near-doubling of monthly churn is **not one event**. It is three factors that overlapped in time:

1. **An app "v3" connected-TV playback regression** (window-month 32) — the acute trigger. Video-start failures tripled on smart-TV / streaming-stick; the heaviest- viewing subscribers lost value and churned a month later. Present in

every market, weakest in the US. This is the single biggest contributor and the reason the company-wide number looks "global".

2. **The Brazil Standard/Premium price increase** (window-month 32) — the amplifier, and **Brazil-only**. It did nothing to engaged subscribers; on subscribers already disengaged — many of them pushed there by Factor 1 — it added ~18 pp of churn. This is why Brazil *steps* rather than drifts.
3. **A months-old Mexico cohort-quality drag** — the pre-existing drain. Mexican subscribers acquired since ~mid-2025 retain ~8 pp worse at month 3; this had already lifted Mexican churn ~0.6 pp *before* window-month 33 and is still underneath the spike. This is why Mexico was the highest market and *ramping* before anyone looked.

No single factor explains it. Remove Factor 1 and Brazil's step is smaller and the US and Mexico barely move; remove Factor 2 and Brazil drifts instead of stepping; remove Factor 3 and Mexico looks fine. In Brazil, **85 % of the elevated churn carries Factor 1 or Factor 2** (only 15 % carries neither); in Mexico, ~64 % carries Factor 1 or Factor 3. The US "38 % carries none" is simply its unchanged ~4 % baseline — the US is the control that shows the mechanism, not a fourth market with its own problem.

Limitations. The three flags overlap (a Brazil CTV-heavy disengaged Std/Prem subscriber carries F1 and F2), so the shares do not sum to a clean attribution; the decomposition establishes *coverage* ("almost none of the excess is unexplained"), not independent percentages. The 52 % F1 share in every market partly reflects the CTV-heavy population share, which is why the **rate** comparisons in `F05` (equal pre-shift, 8.4 % vs 5.6 % at peak) carry the causal weight, not the flag share alone. Data is synthetic and seed-generated (`docs/08`).

Recommendation (priority order).

1. **Fix the connected-TV player** (Sev-1). Roll back / hotfix v3 on smart-TV and streaming-stick; verify video-start-failure returns to ~1 %. Save-campaign the CTV-heavy subscribers whose viewing dropped after window-month 32.
2. **Pause the Brazil price migration for low-engagement Standard/Premium subscribers**; give them a retention path (hold price, downgrade, or pause) and re-test elasticity after the CTV fix.
3. **Audit Mexico acquisition and onboarding** since mid-2025; cap spend on the weakest sources until month-3 retention recovers toward ~82 %.
4. **Instrument** so this is visible next time: connected-TV QoE on the exec dashboard, a leading "engagement health" trend (`F04`), and price-change churn elasticity by engagement band.

that produced the spike is not stated in any document; it lives only in `etl/config.py` and was recovered from the data. Per-finding method and evidence: `data_analysis/findings/F01 – F10` .